

CompTIA.

A+ Core 1

Exam 220-1201

A+ 220-1201

Module 10



Supporting Print Devices

Learning Outcomes

Prepare for A+ Core 1 by:

- Deploying printer and multifunction devices
- Replacing print device consumables
- Troubleshooting print device issues

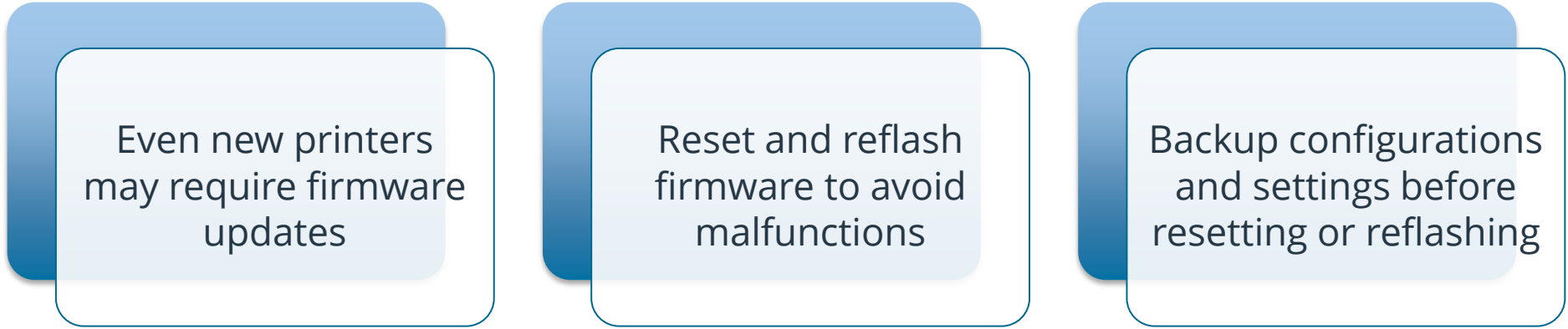
Lesson 10.1

Printers and Multifunction Devices

Printer Unboxing and Setup Location

- Select a printer
 - Consider speed, resolution, paper handling, options
- Setup location
 - Consider power, network, environmental concerns, accessibility
- Unboxing a printer
 - Lifting heavier units may require two people
 - Packing materials
 - Acclimation to humidity and temperature

Firmware Management in MFDs and Printers

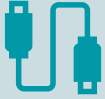


Even new printers
may require firmware
updates

Reset and reflash
firmware to avoid
malfunctions

Backup configurations
and settings before
resetting or reflashing

Print Device Connectivity (Slide 1 of 5)



USB is directly connected to a computer

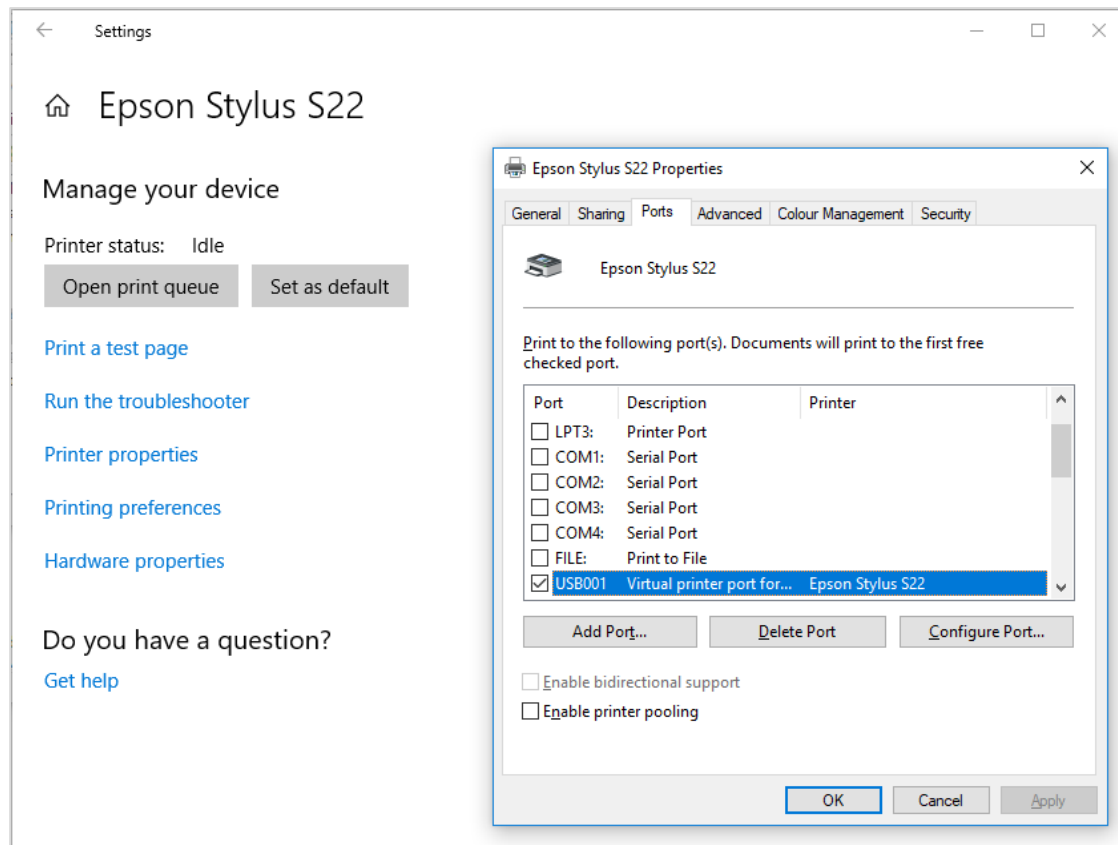


Ethernet is connected to network with its own IP address



Wireless is connected to Wi-Fi network or to computer via Bluetooth

Print Device Connectivity (Slide 2 of 5)



Print Device Connectivity (Slide 3 of 5)

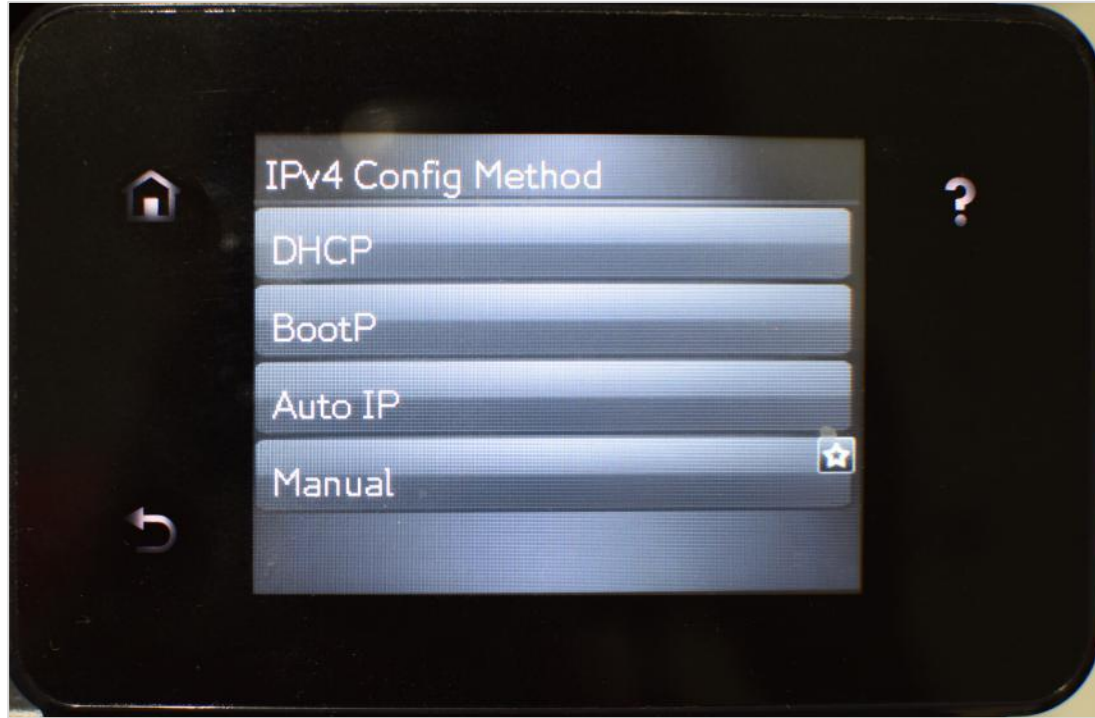
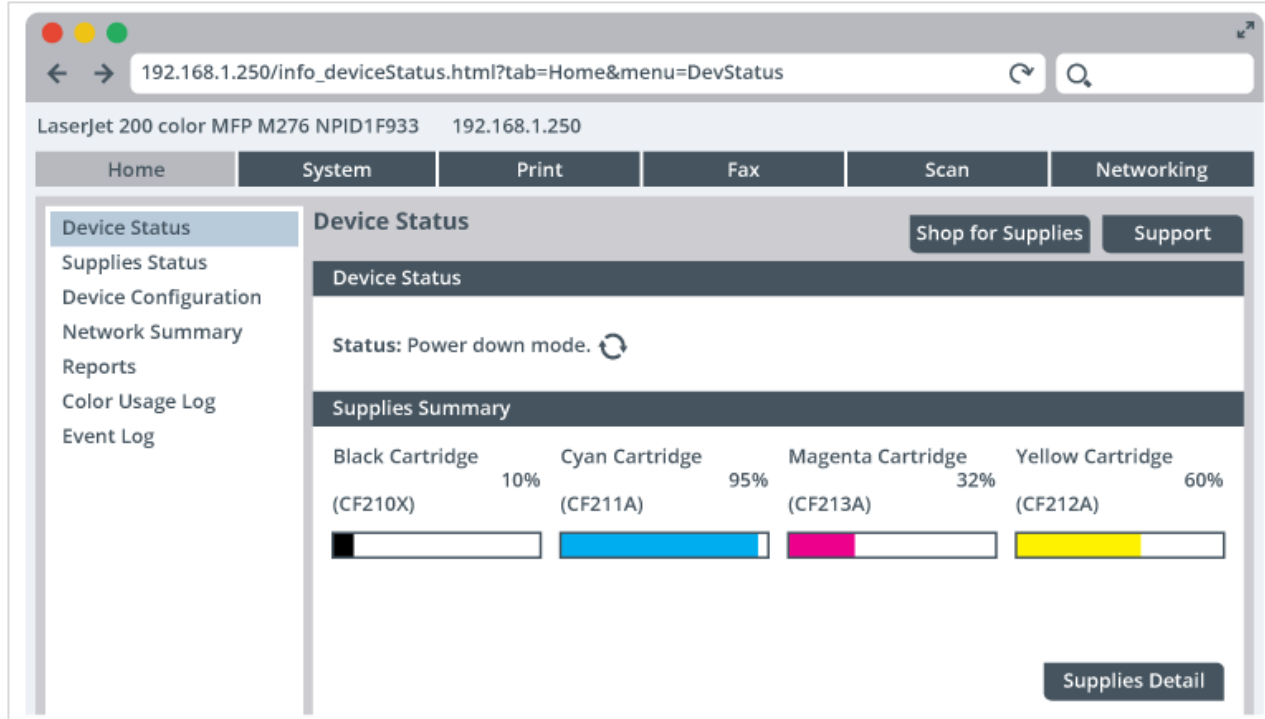


Image courtesy of CompTIA.

Print Device Connectivity (Slide 4 of 5)



Print Device Connectivity (Slide 5 of 5)



Image courtesy of CompTIA.

Printer Drivers and Page Description Languages (Slide 1 of 2)

- Driver Selection
 - Plug and play
 - Manual installation for page descriptive language (PDL)
 - XPS, PCL, and PostScript
- PDL supports
 - Scalable fonts
 - Vector graphics
 - Color printing
 - CMYK- Cyan, Magenta, Yellow, Black

Printer Drivers and Page Description Languages (Slide 2 of 2)

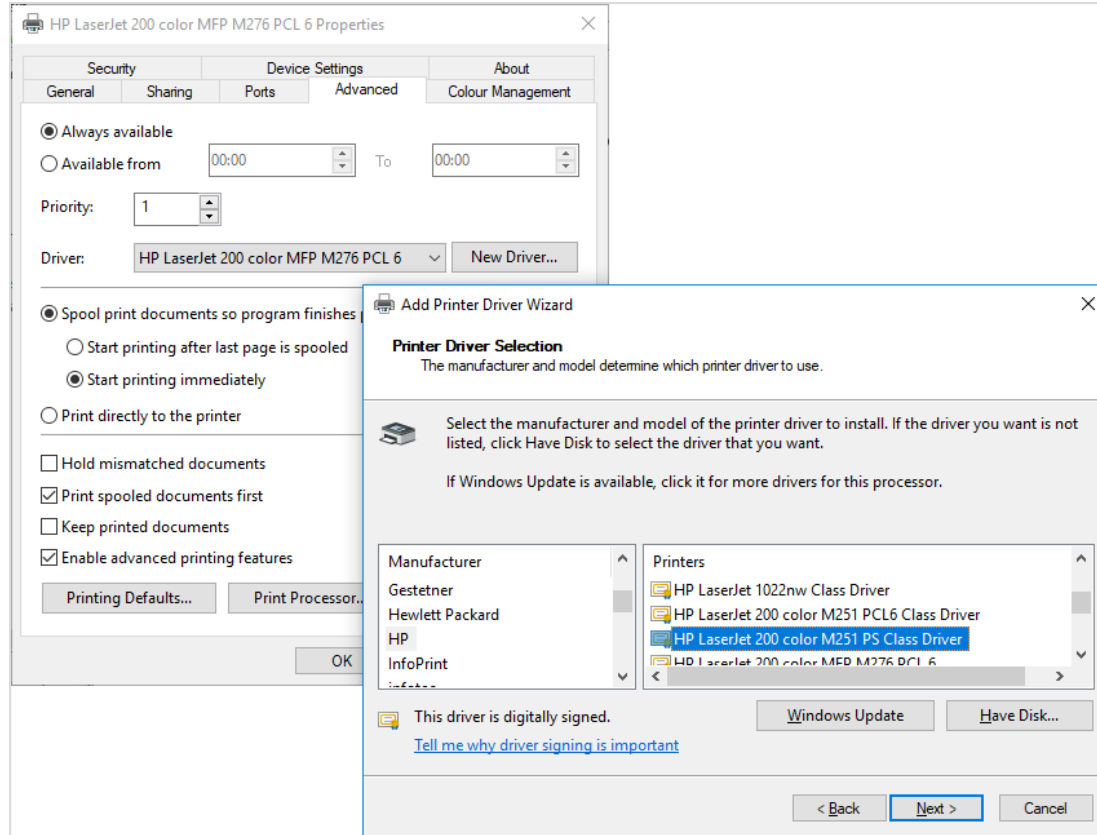
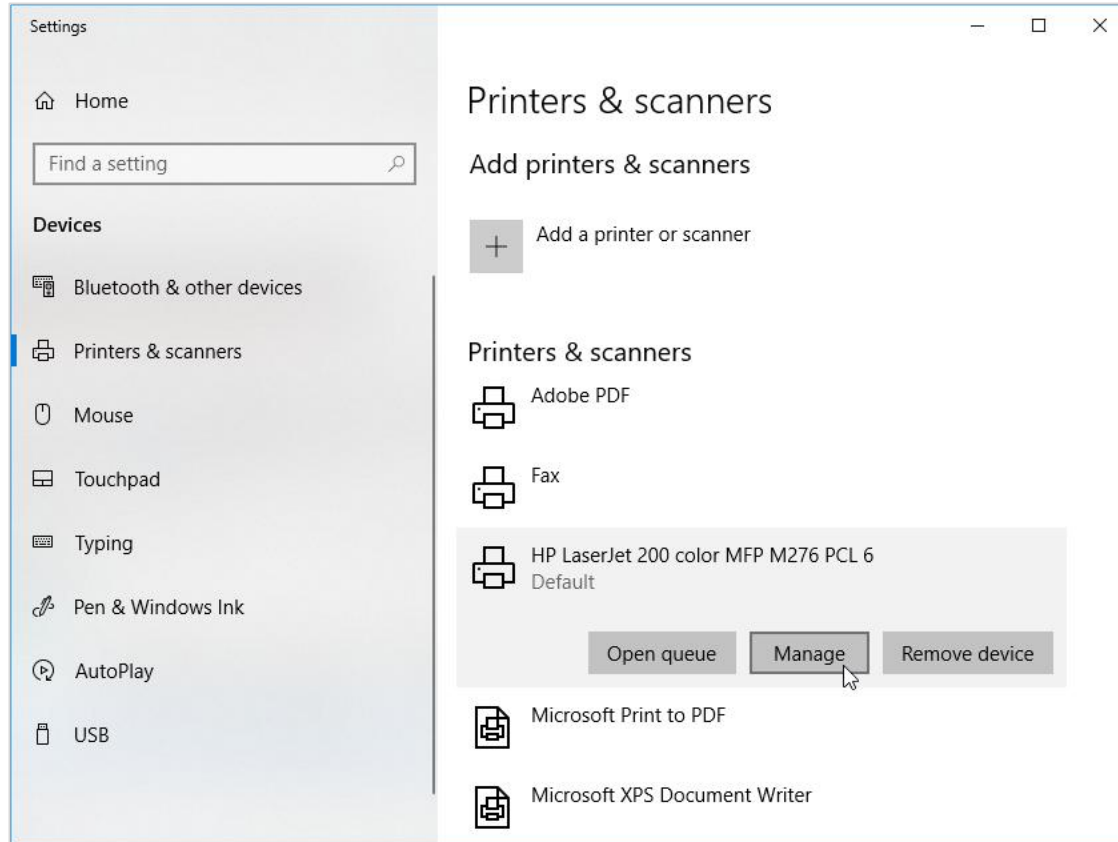


Image courtesy of Microsoft.

Printer Properties (Slide 1 of 3)

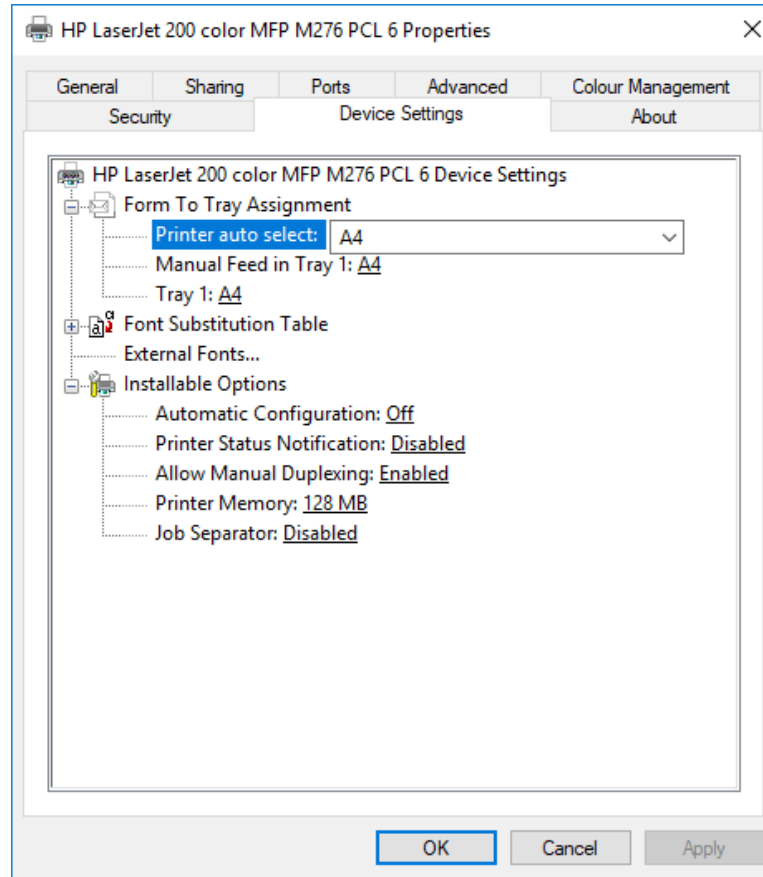
- Print queue
- Paper tray selection
- Fonts
- Driver

Printer Properties (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Printer Properties (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Printing Preferences (Slide 1 of 3)



Print quality

DPI (dots per inch)
economy/draft mode



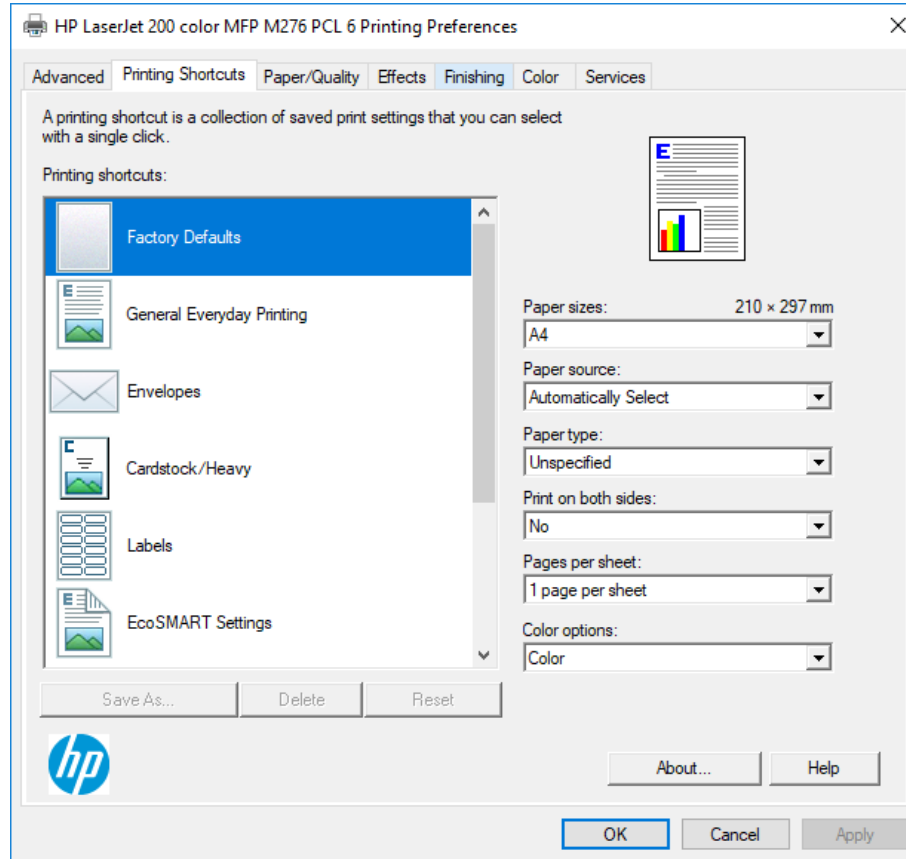
Paper type and size



Finishing

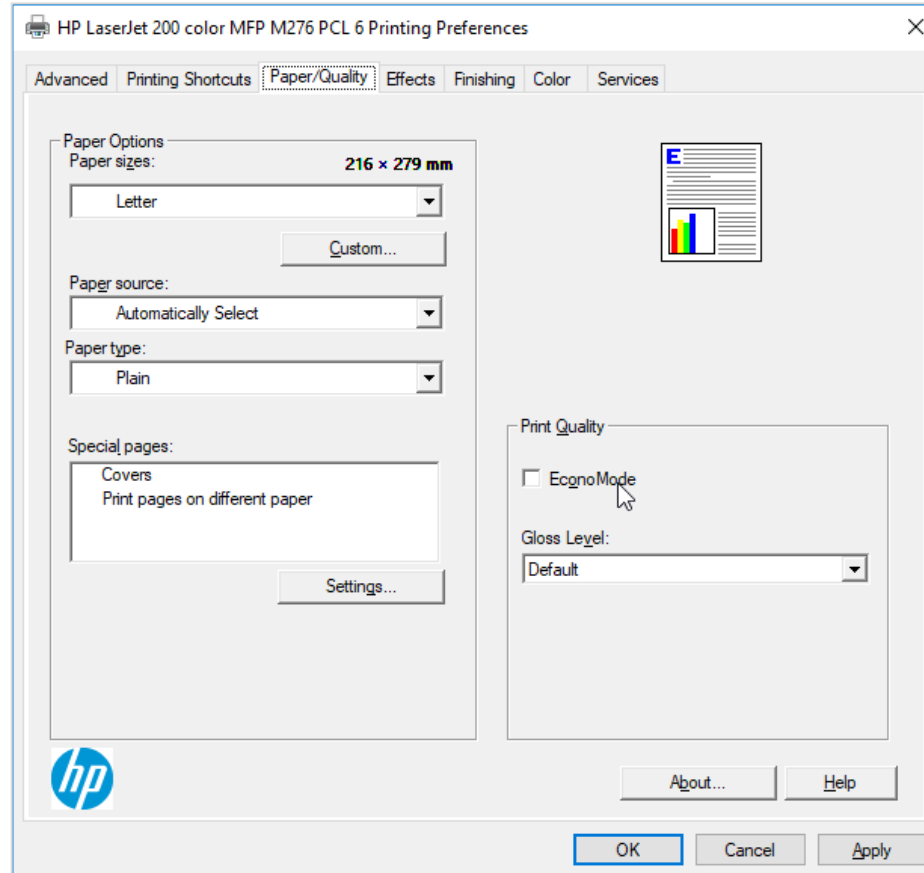
Duplex
Multiple pages per sheet
Landscape vs. portrait

Printing Preferences (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Printing Preferences (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Printer Sharing (Slide 1 of 4)



Public printers have no access controls

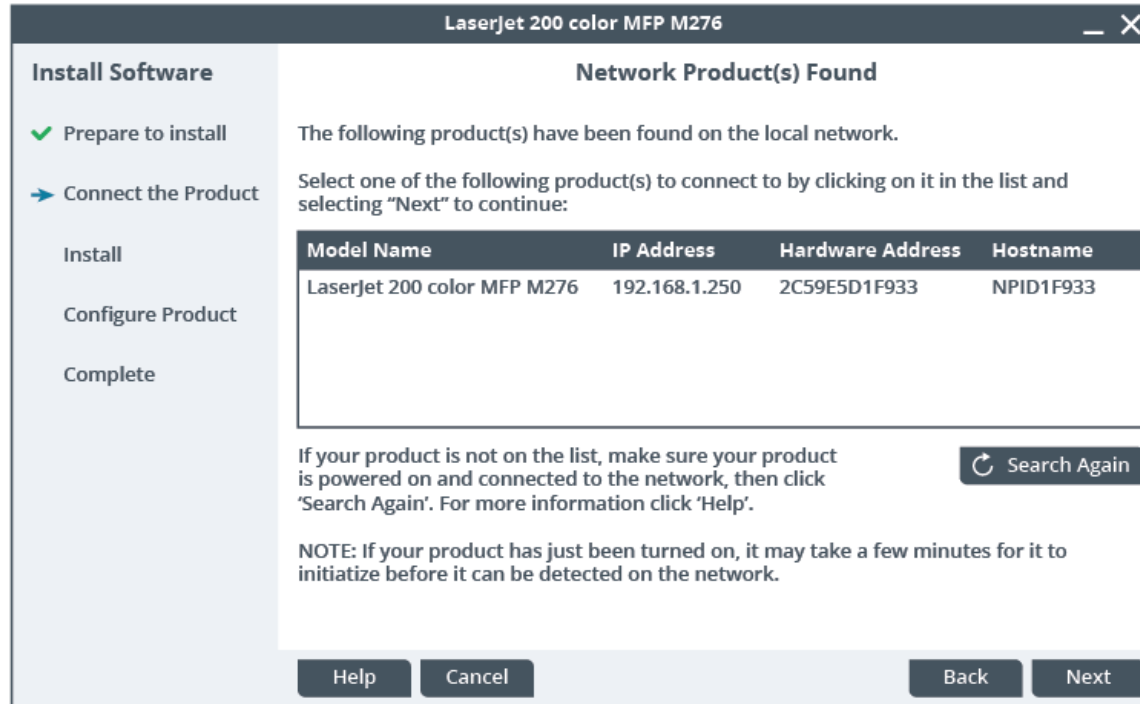


Print server are connected to server/computer then shared to network

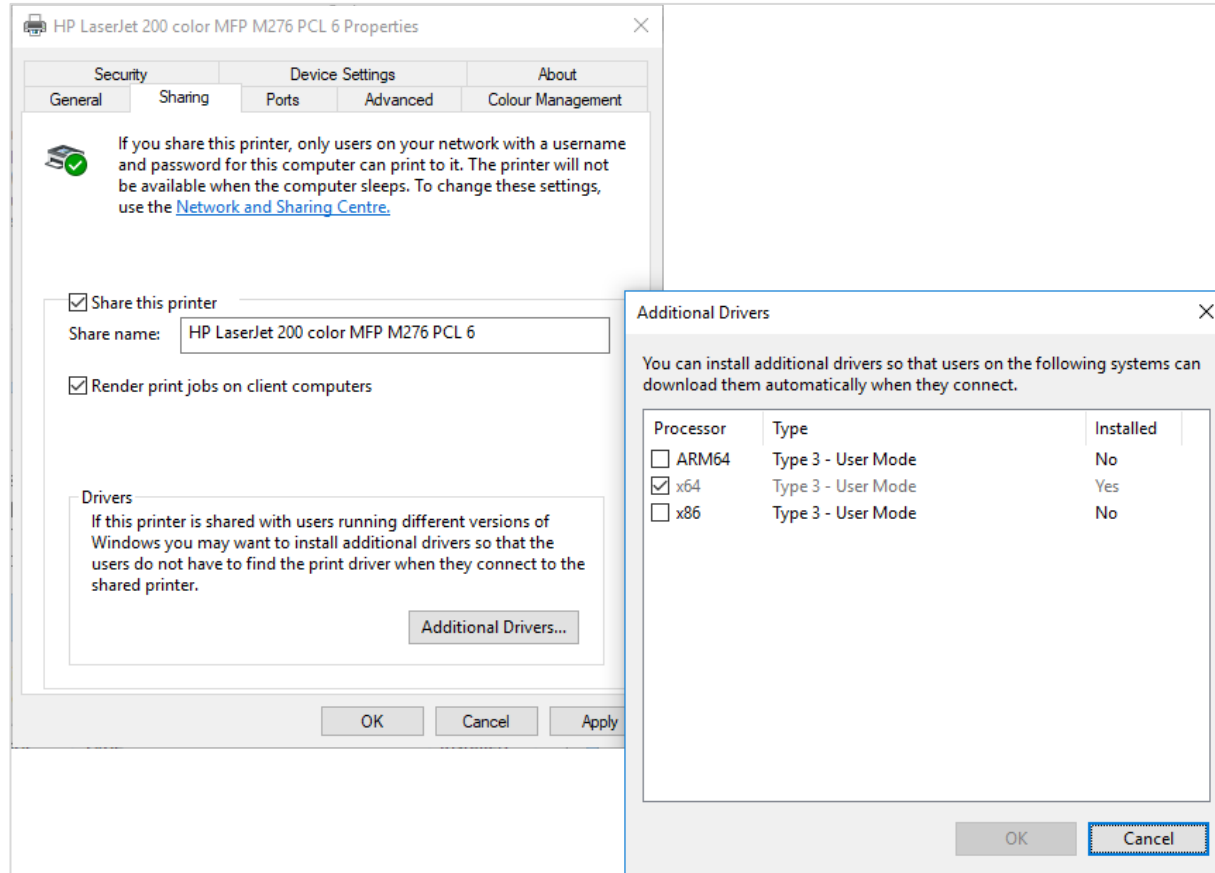


Shared printer connections

Printer Sharing (Slide 2 of 4)

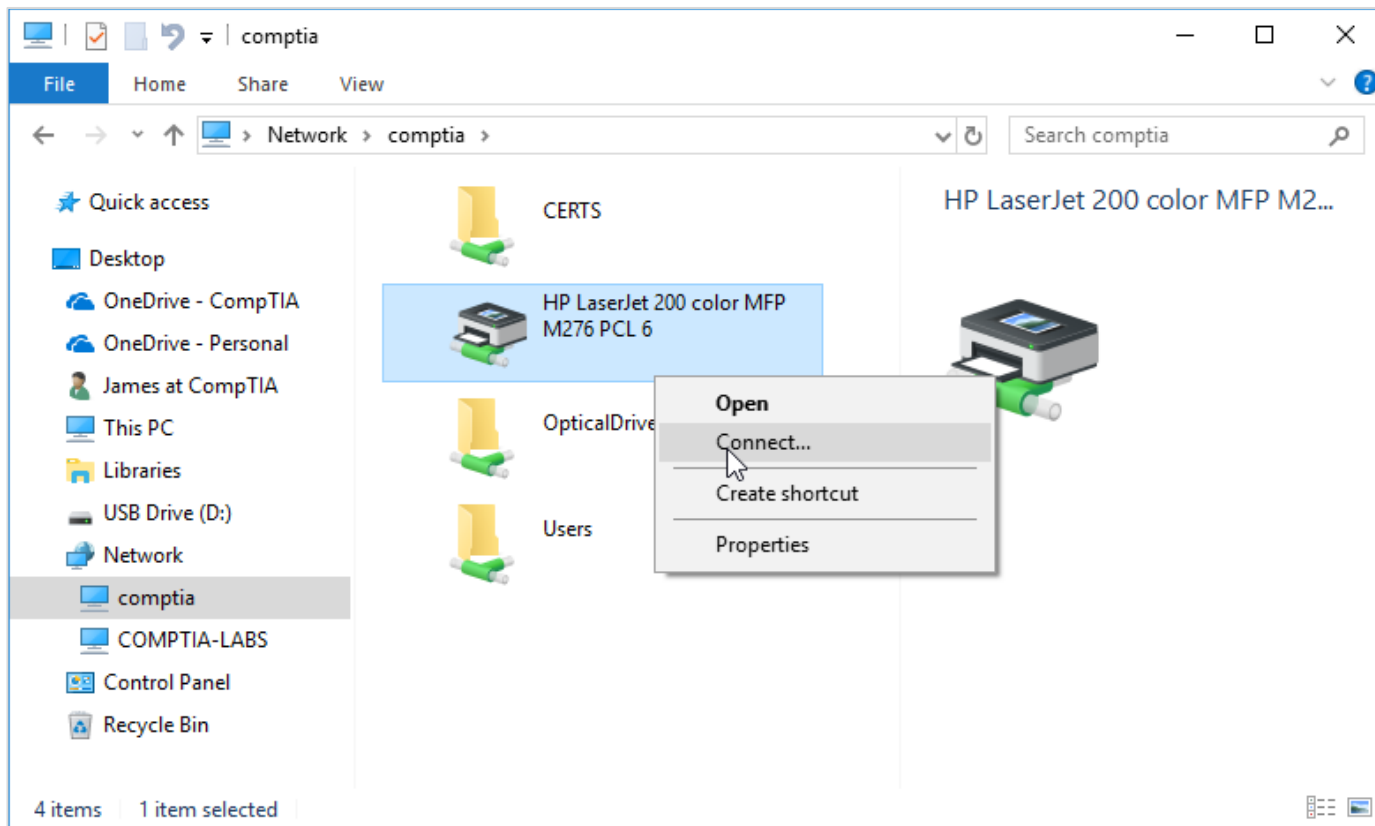


Printer Sharing (Slide 3 of 4)



Screenshot courtesy of Microsoft.

Printer Sharing (Slide 4 of 4)



Screenshot courtesy of Microsoft.

Printer Security



USER AUTHENTICATION



PIN ENTRY OR BADGE SWIPE
CAN RELEASE PRINT JOBS



AUDIT LOGS TRACK DETAILS
OF EACH PRINT JOB

Scanner Configuration (Slide 1 of 2)

- Optical Character Recognition (OCR) identifies characters to recreate document
- Flatbeds use moving head across document
- Automatic Document Feeder (ADF) uses fixed scan head
- Network scanning services
 - Email
 - Server Message Block (SMB)
 - Cloud

Scanner Configuration (Slide 2 of 2)

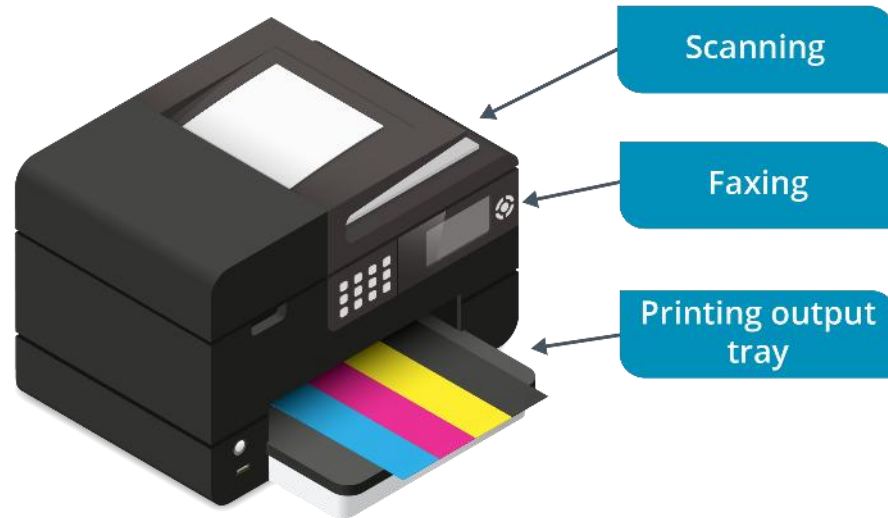


Image © 123RF.com

Lab Activity

- Lab: Select and Install a Printer
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure Network Printing
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Printers and Multifunction Devices

- Printer Unboxing and Setup Location
- Firmware Management in MFDs and Printers
- Print Device Connectivity
- Printer Drivers and Page Description Languages
- Printer Properties
- Printing Preferences
- Printer Sharing
- Printer Security
- Scanner Configuration

Lesson 10.2

Print Device Maintenance

Laser Printer Imaging Process (Slide 1 of 3)

- Step 1 Processing
 - Document is encoded and sent to printer to be converted to bitmap
- Step 2 Charging
 - Charge roller applies negative charge to imaging drum
- Step 3 Exposing
 - Laser shines on imaging drum, removing the negative charge
- Step 4 Developing
 - Toner sticks to neutralized areas on imaging drum

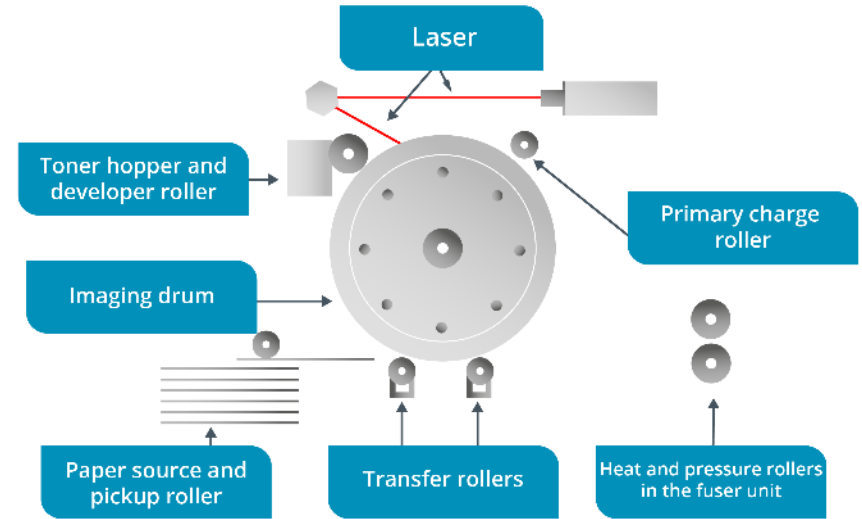


Image © 123RF.com.

Laser Printer Imaging Process (Slide 2 of 3)

- Step 5 Transferring
 - Toner is transferred to paper
- Step 6 Fusing
 - Heat is applied to fuse toner to paper
- Step 7 Cleaning
 - Imaging drum is cleaned of residual toner

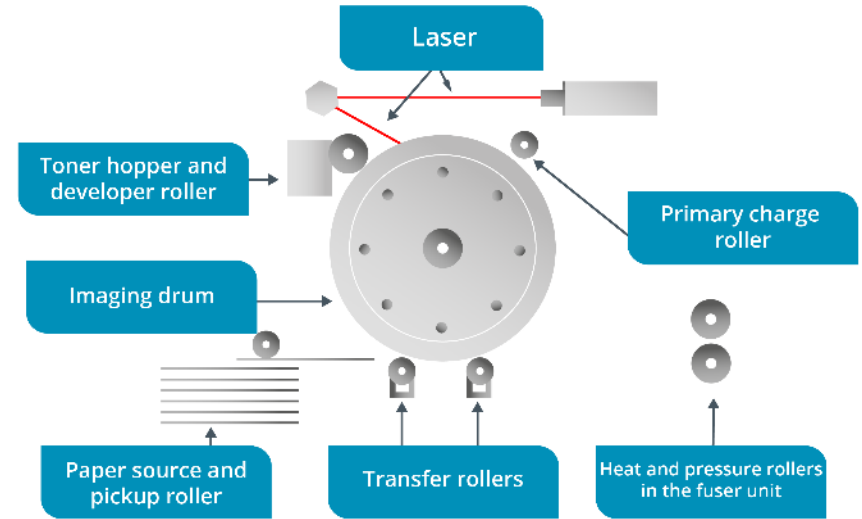


Image © 123RF.com.

Laser Printer Imaging Process (Slide 3 of 3)



Image courtesy of CompTIA.

Laser Printer Maintenance

Load paper when empty

Replace toner cartridge

- Ensure packing is removed before installation

Clean printer

- Never use compressed air
- Use damp cloth to wipe exterior

Maintenance Kit

- Fuser/imaging unit
- Rollers

Calibration

Inkjet Printer Imaging Process (Slide 1 of 2)

- Thermal printers
 - Heat ink to form a bubble that bursts, spraying ink
- Piezoelectric printers
 - Use jet nozzle to spray ink
- Carriage system
 - Holds print head and sometimes ink cartridges

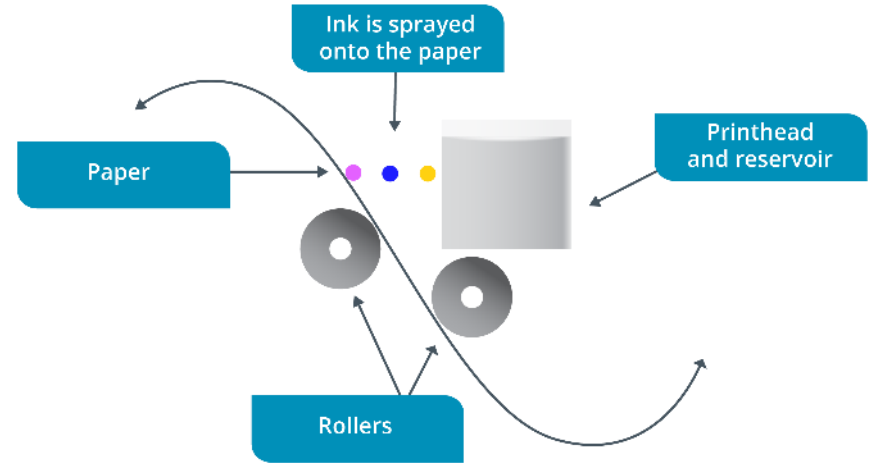


Image © 123RF.com.

Inkjet Printer Imaging Process (Slide 2 of 2)



Image by Erik Bobeldijk © 123RF.com

Inkjet Printer Maintenance (Slide 1 of 3)

- Replace rollers regularly
- Ensure paper is installed correctly
- Replace inkjet cartridges
- Printers with ink tanks are refilled with liquid ink
- Calibrate print head alignment
- Clean print head
- Fix paper jams

Inkjet Printer Maintenance (Slide 2 of 3)

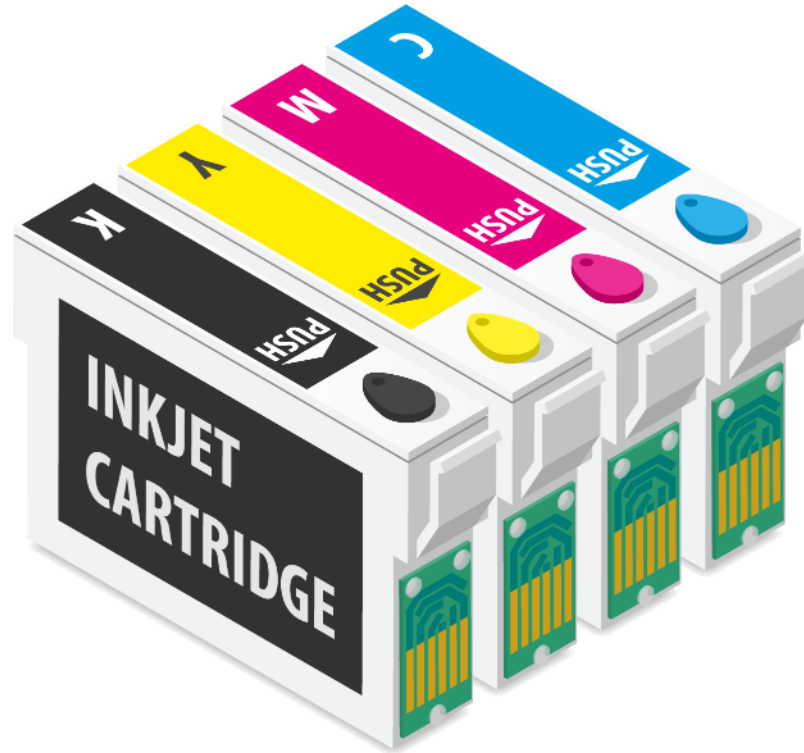
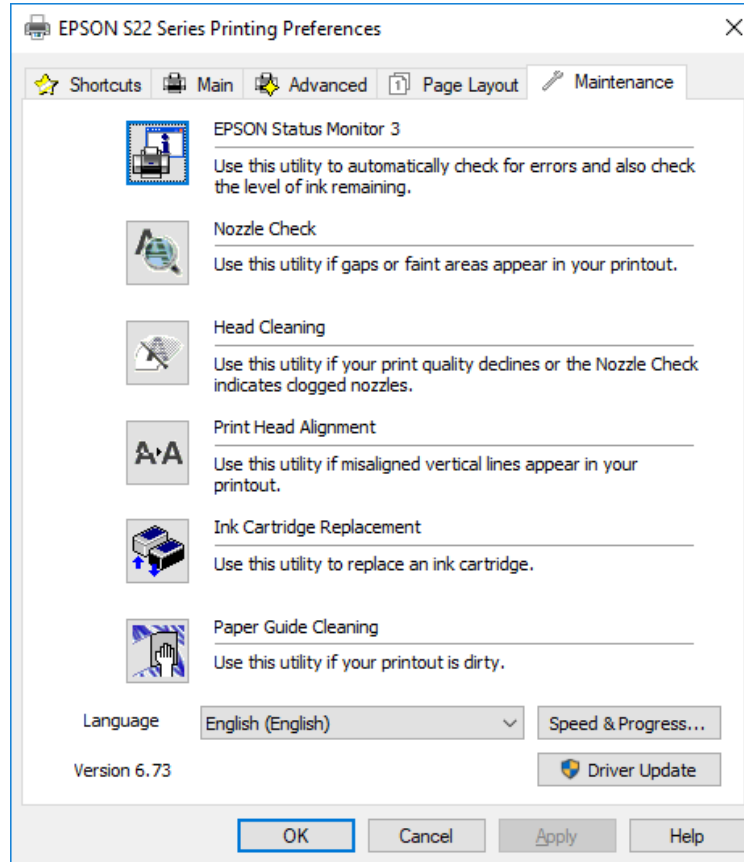


Image © 123RF.com

Inkjet Printer Maintenance (Slide 3 of 3)



Screenshot courtesy of Microsoft.

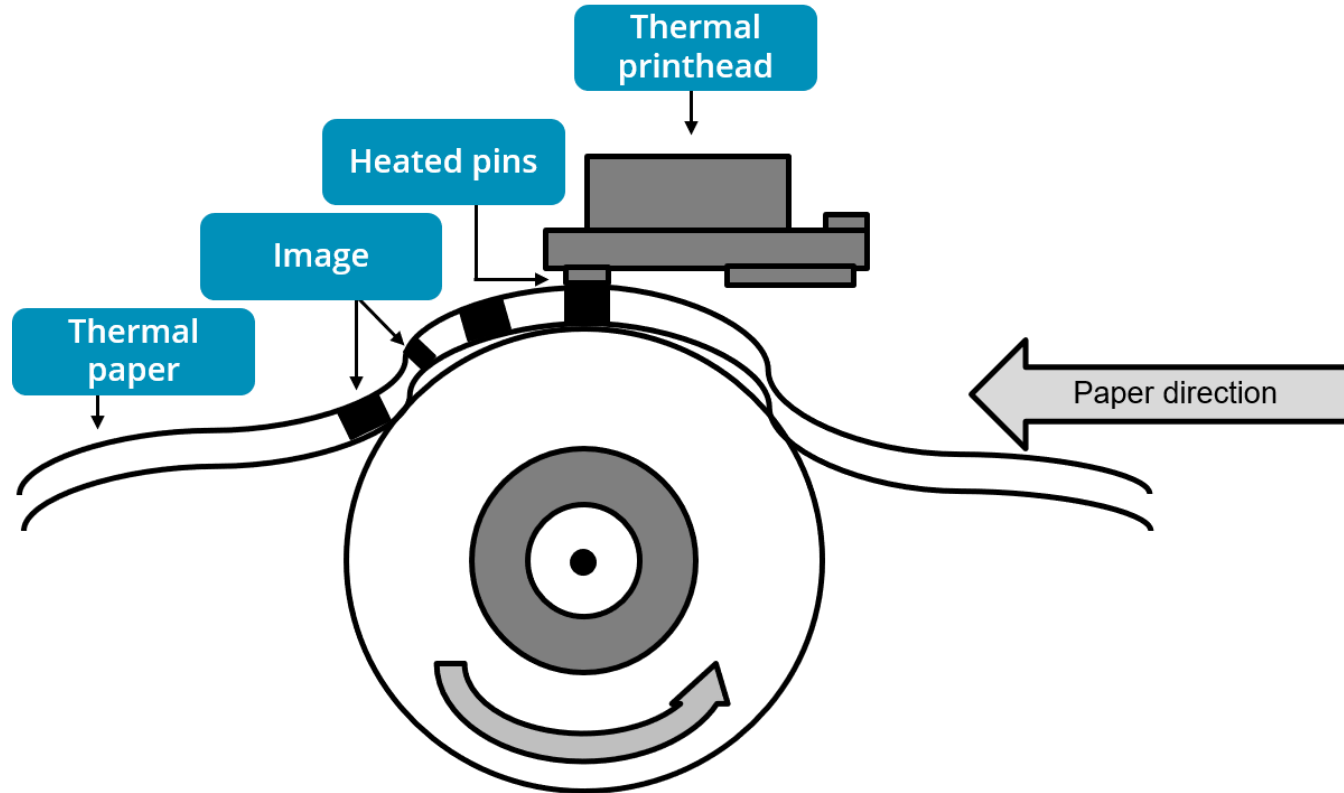
Thermal Printer Maintenance (Slide 1 of 2)

- Wax-coated paper heats to transfer image
- Commonly for receipts, shipping labels, and barcode labels
- Tearing creates dust so clean regularly



Image © 123RF.com.

Thermal Printer Maintenance (Slide 2 of 2)



Impact Printer Maintenance

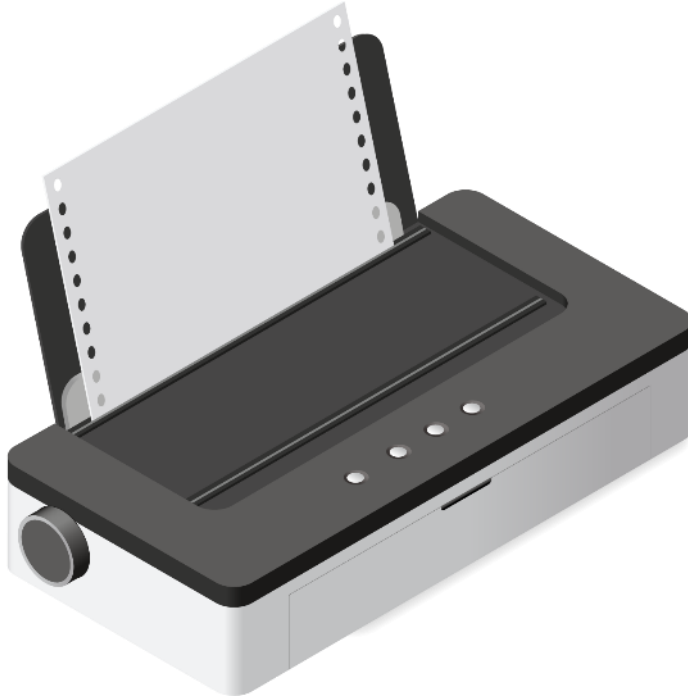


Image © 123RF.com

- A dot matrix is a series of dots used to form characters
- Multipart paper is carbon/carbonless paper with white, yellow, and pink layers
- Create multiple copies in one pass
- Tractor-fed paper has holes on side to aid feeding

Lab Activity

- Replace a Printer Fuser
(available in CertMaster Learn and CertMaster Perform)
- Replace a Printer Toner
(available in CertMaster Learn and CertMaster Perform)
- Clear a Paper Jam in a Printer
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Print Device Maintenance

- Laser Printer Imaging Process
- Laser Printer Maintenance
- Inkjet Printer Imaging Process
- Inkjet Printer Maintenance
- Thermal Printer Maintenance
- Impact Printer Maintenance

Lesson 10.3

Troubleshoot Print Devices

Printer Connectivity Issues

- Check if printer is powered on and online
- Verify consumables (ink, paper) are loaded
- Print a test page
- Inspect cables or wireless settings
- Check for needed firmware and driver updates
- For cloud printers, verify registration with cloud service

Print Feed Issues

- For paper jams
 - Follow on-screen instructions to clear paper path
- Verify paper settings and feed roller condition
- Grinding noises
 - Ensure cartridges and print head are seated correctly



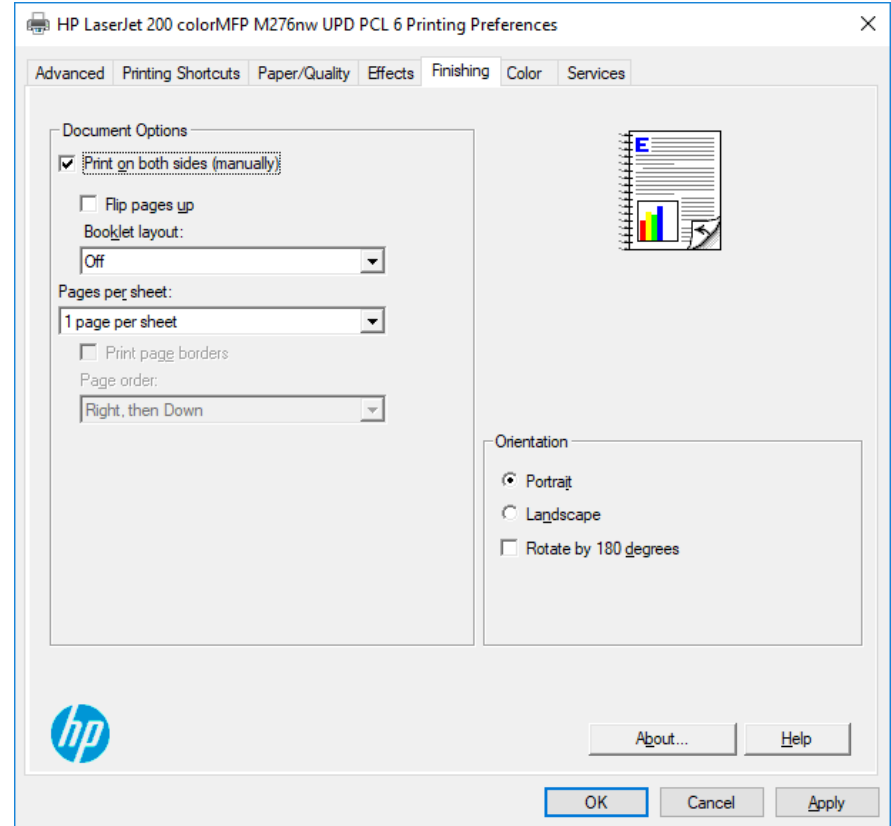
Image courtesy of CompTIA.

Print Quality Issues

- Faded or faint
 - Verify print resolution and/or replace toner cartridge
- Black stripes
 - Check charge roller or high-voltage power supply
- Vertical or horizontal lines
 - Clean rollers
- Smudged toner
 - Replace fuser
- A color is missing
 - Replace cartridge and clean contacts
- Print test pages to verify the results

Finishing Issues

- Check for incorrect page orientation
- Hole punch errors
 - Verify max number of pages allowed
- Staple errors
 - Check for a stapler jam
 - Check for staples empty



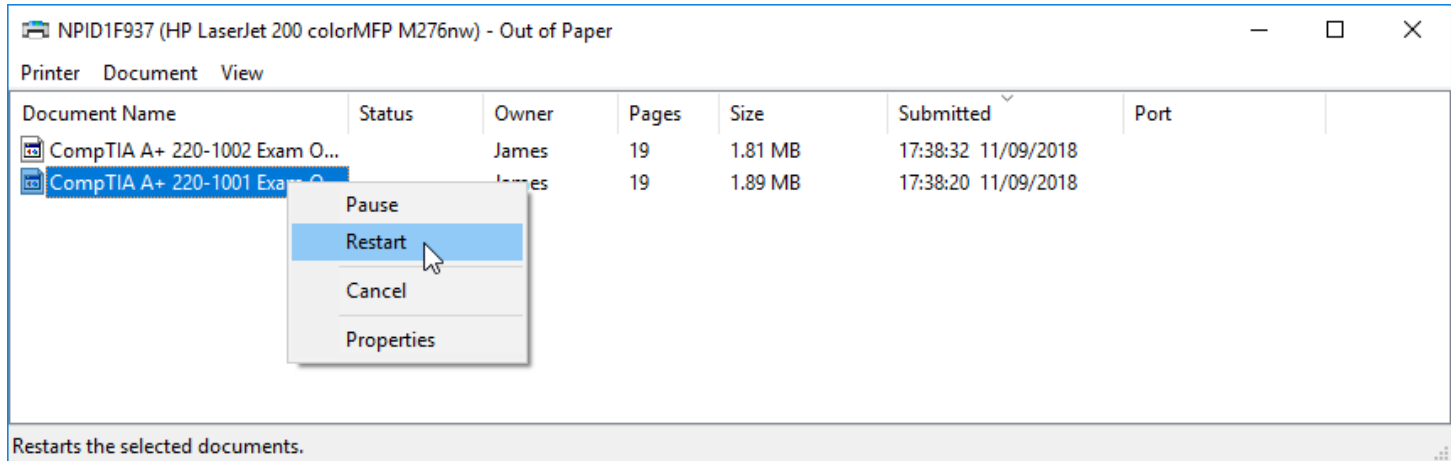
Screenshot courtesy of Microsoft.

Print Job Issues (Slide 1 of 2)

- Print monitor provides printing status updates
- Clear old or obsolete jobs from print queue and spooler
- Tray not recognized error
 - Verify tray is fully inserted
- Frozen print queue
 - Stop and restart print spooler and clear cache

Print Job Issues (Slide 2 of 2)

- Garbled print job
 - Verify correct PDL is selected
 - Print test page
 - If test page is good, it may be an application-specific issue



Screenshot courtesy of Microsoft.

Lab Activity

- Lab: Resolve Print Services Support Tickets
(available in CertMaster Learn and CertMaster Perform)



Lesson Review: Troubleshoot Print Devices

- Printer Connectivity Issues
- Print Feed Issues
- Print Quality Issues
- Finishing Issues
- Print Job Issues